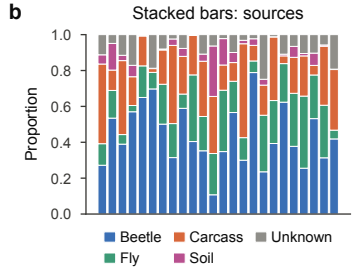
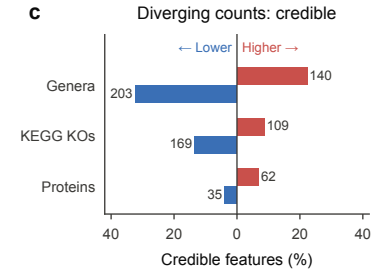
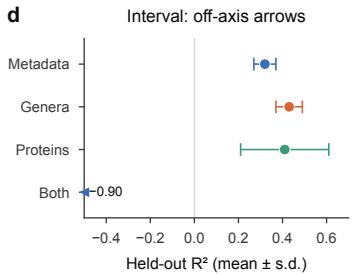
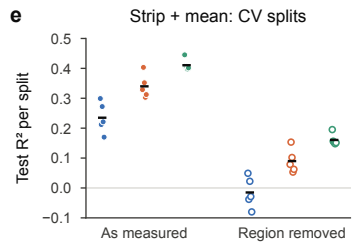
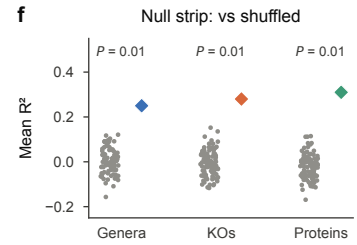

a
Heatmap: leave-one-out
Beetle 0.22 0.32 0.27 0.08 0.11
Fly 0.29 0.00 0.28 0.27 0.16
Carcass 0.17 0.16 0.14 0.25 0.28
Soil 0.14 0.25 0.20 0.16 0.25
Beetle Fly Carcass Soil Unknown
Mean proportion
1.0
0.5
0.0

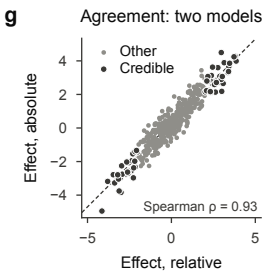

b
Stacked bars: sources
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
Beetle
Carcass
Unknown
Fly
Soil

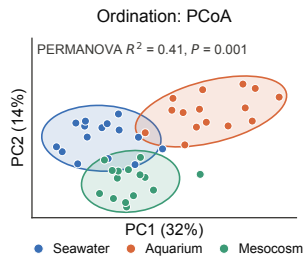

c
Diverging counts: credible
← Lower
Higher →
Genera
KEGG KOs
Proteins
203
140
169
109
35
62
40
20
0
20
40
Credible features (%)

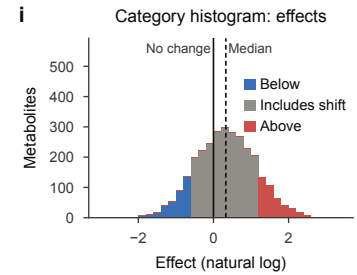

d
Interval: off-axis arrows
Metadata
Genera
Proteins
Both
0.90
−0.4
−0.2
0.0
0.2
0.4
0.6
Held-out R² (mean ± s.d.)

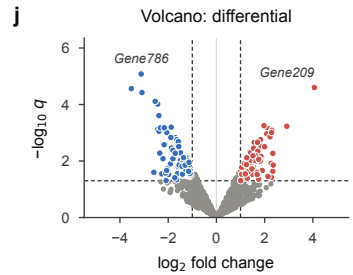

e
Strip + mean: CV splits
Test R² per split
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
As measured
Region removed

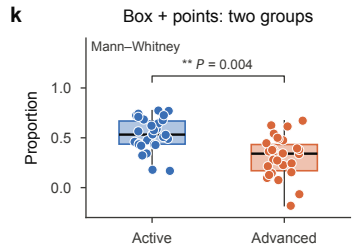

f
Null strip: vs shuffled
Mean R²
P = 0.01
P = 0.01
P = 0.01
0.4
0.2
0.0
−0.2
Genera
KOs
Proteins

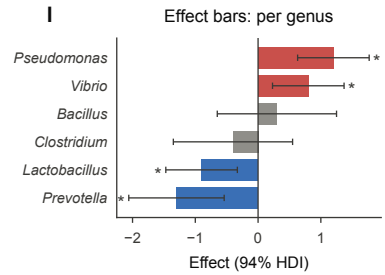

g
Agreement: two models
Other
Credible
Effect, absolute
4
2
0
−2
−4
Spearman ρ = 0.93
−5
0
5
Effect, relative

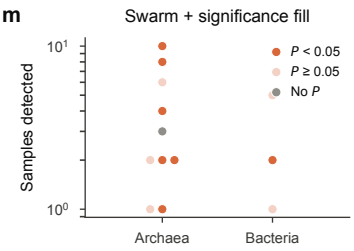

h
Ordination: PCoA
PERMANOVA R² = 0.41, P = 0.001
PC2 (14%)
PC1 (32%)
Seawater
Aquarium
Mesocosm


i
Category histogram: effects
No change
Median
Below
Includes shift
Above
Metabolites
500
400
300
200
100
0
−2
0
2
Effect (natural log)


j
Volcano: differential
Gene786
Gene209
−log10 q
6
4
2
0
−4
−2
0
2
4
log2 fold change


k
Box + points: two groups
Mann–Whitney
** P = 0.004
Proportion
1.0
0.5
0.0
Active
Advanced


l
Effect bars: per genus
Pseudomonas
Vibrio
Bacillus
Clostridium
Lactobacillus
Prevotella
−2
−1
0
1
Effect (94% HDI)


m
Swarm + significance fill
Samples detected
10¹
10⁰
P < 0.05
P ≥ 0.05
No P
Archaea
Bacteria